

ELLEN BRISLEY

EU Citizen | ecbriz26@outlook.com | [LinkedIn](#) | [Portfolio](#)

SUMMARY

- Strong hardware and physics experience, combined with demonstrated leadership in translating complex technical concepts across cross-functional and non-technical teams.
- **Leadership:** Canada's largest undergraduate merit scholarship, \$100,000
- **Technical tools:** KiCad, soldering, micro-controllers (ESP32, Arduino, Raspberry Pico), lab measurement tools, Computer-Aided Design (Solidworks, OnShape), Python, Java.
- **Languages:** Fluent in English and French (spoken, reading, writing). French, Canadian, UK citizen.

EDUCATION

University of British Columbia | Engineering Physics (BASc) | 91% Cumulative Average | 4th Year

WORK EXPERIENCE

Electrical Engineering Intern | CERN, ISOLTRAP Experiment, Geneva Sept 2026 - Dec 2026

- Currently designing high voltage RF DC Mixing Board.

Electrical Engineering Intern | Max Planck Institute For Nuclear Physics, Germany May 2026 - Sept 2026

- **Multi-iteration circuit and PCB design** of low signal X-ray detector; to **replace €20k detector**.
- Designed 4-wire resistance and capacitive measurement PCBs for liquid helium and liquid nitrogen level sensing inside a superconducting magnet; cut ordering lead times and **replaced €6k in commercial instrumentation**.
- Redesigned vacuum-protection safety circuit; compact layout and box to fit tightly behind pump.

Water Filter Engineering Intern | [Clear Cambodia](#), Phnom Penh Jan 2025 - May 2025

- **Designed and developed prototype** of compact BioSand Filter - [see Blog](#)
- Led **2-month field experiment** in Cambodia to evaluate filter efficiency; demonstrated filtration performance on par with control filters.
- Worked closely with Cambodian staff; navigated **cross-cultural and multilingual teamwork**.
- **Principal author and presenter** at UNC Water & Health Conference.

Research Assistant | TRIUMF Particle Accelerator TITAN Ion Trap May 2024 - Sept 2024

- **Led Python data analysis** on neutron-deficient Cs/Ba isotopes and isomers.
- Participated in experiments, gathered data, kept scientific logs, and analyzed data on site.
- Third place in TRIUMF undergraduate [poster presentation](#), among senior students.

TECHNICAL PROJECTS

Autonomous Robot Design | UBC Engineering Physics May 2025 - Aug 2025

- Strong performance in ([final competition](#)), autonomously navigated surface and rescued 5 out of 7 objects.
- Designed and prototyped claw and linear actuator system with CAD and laser/waterjet cutting.
- Troubleshoot circuits integrating ESP32, Hall-Effect/LiDAR, rotary encoders, and stepper/DC motors.
- Collaborated in 4-person team; delivered a fully functional robot within 6-week sprint.

Social App Design | Aiuto Solutions Oct 2020 - Aug 2023

- Designed, coded, pitched a self-made donation transparency platform; used to provide 200+ personalized gifts to nursing home residents during the pandemic.
- Recognized by Mayor ([LINK](#)), Member of Parliament, CBC News ([LINK](#)).

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AWARDS & LEADERSHIP

- **Loran Scholar** | \$100,000 Undergraduate Merit Award March 2023 - Present
 - Selected as 1 of 36 scholars from 4,800 applicants through competitive national interviews.
 - Recognized for leadership, character, service, and academics; award funds 4 years of tuition, living expenses, and experiential work terms in community development, policy, and enterprise.
 - Supplemented by \$3,000 in additional UBC academic scholarships.
- **Erich Vogt First Year Summer Research Experience (FYSRE)** May 2024
 - Selected as 1 of 3 first-year UBC students for a paid research placement at TRIUMF Particle Accelerator.
- **Canadian Governor General's Academic Medal** June 2023
 - Awarded for highest academic average at Cameron Heights Collegiate Institute (IB: 43/45).
- **Technovation World Summit Awards** | \$1000 scholarships Aug 2019, Aug 2021
 - Economic Award and North America Regional Winner in global tech entrepreneurship competition.

PUBLICATIONS

- **Principal author:** "Compact BioSand Filter Prototype Performance in Cambodia: Field Comparison with a Full-Sized Plastic BSF" - verbal presentation at UNC Water & Health Conference.
- **Co-author:** "Formation of gaseous, doubly charged cerium monofluoride CeF_2^{2+} and its sensitivity to new physics" - in review for Physical Review X Journal.